Lesson Planning

Complete Lesson Plan Book

Class: FSc Pre-Medical · Class 12

Subject: Chemistry

Medium: English

Academic Year 2026-27

DM

This book is prepared as a teaching guide for teachers.

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Unit 1: Acids, Bases and the pH Scale

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Learning Objectives

• Students will classify solutions as acidic, basic or neutral using the pH scale.

• Students will explain the difference between strong and weak acids and bases.

• Students will write the general word equation for a neutralisation reaction.

Key Concepts

• An acid donates hydrogen ions in water, while a base accepts them or releases hydroxide ions.

• The pH scale runs from 0 to 14: below 7 is acidic, 7 is neutral, and above 7 is basic or alkaline.

• pH is a logarithmic scale, so a change of one pH unit means a tenfold change in hydrogen ion concentration.

• Strong acids and bases ionise almost completely, while weak ones ionise only partly.

• Neutralisation is the reaction of an acid with a base to form a salt and water.

Teaching Methodology (Step by Step)

1. Ask students why lemon juice tastes sour and soap feels slippery, and link both to acids and bases.

2. Draw the pH scale from 0 to 14 and mark common liquids on it with the class.

3. Explain hydrogen ions, strong and weak ionisation, and the logarithmic meaning of pH.

4. Students test three solutions with litmus paper and record their results.

5. Students write the neutralisation equation and give one everyday example.

Classroom Activities

• pH sorting: students place everyday liquids such as lemon juice, soap solution and pure water on a pH scale strip.

• Colour change: students test solutions with litmus paper and record acid or base results in a table.

• Neutralisation demo: the teacher adds an acid to a base drop by drop and students note the pH change.

Assessment Questions

1. Is a solution of pH 3 acidic or basic? Give a reason.

2. What is the pH of a neutral solution at room temperature?

3. Name one strong acid and one weak base, and state one use of each.

Homework

• List three acids and three bases found at home, with one use of each.

• Explain why toothpaste is slightly basic, in three lines.

Time Allocation

40 minutes